

Harshita Mhaske

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SUMMARY

Robotics Software Engineer with hands-on experience in legged locomotion, reinforcement learning, embedded systems, state estimation, computer vision, path planning, and ROS-based deployment. Skilled in developing and testing robotic systems in both simulation and real-world environments.

EDUCATION

California State University, Fullerton,

MS in Computer Engineering

(Aug 2023 - May 2025)

Graduate Thesis - Legged Robotics under Dr Jidong Huang

SKILLS

Languages & Tools: Python, C++, Git, Docker, Linux

Frameworks & Libraries: ROS, PyBullet, Gazebo, OpenCV, Numpy, TensorFlow, PyTorch, Scikit-learn, Pandas

Hardware: FD-CAN, I2C, IMU, FSR, 3D Printing

WORK EXPERIENCE

Graduate Research Assistant

(Sep 2024 - May 2025)

Institute of Navigation (ION) Lab, Advisor – Dr. Yu Bai, Dr. Doina Bein

- Built an open source low-cost (~\$1000) 12 DOF 3D-printed **quadruped robot** capable of walking on uneven terrain
- Integrated **FSRs and IMU** to feed sensory data to linear policy and tested successfully on uneven ground
- Mentored 100+ underrepresented students** in C++, Python, machine learning, and robotics using Raspbot robotics kits during ACCESS workshops

NASA L'SPACE PROGRAM, Arizona State University

(Mar 2024 - Aug 2024)

- Led the design of an autonomous docking interface between a Mars rover (UGV) and aerial drone (UAV)
- Developed a proposal to enable power transfer and path planning with UAV guidance to optimize rover traversal

Robotics Software Engineer

(Feb 2023 - June 2023)

ARTPARK, Indian Institute of Science, Bangalore

- Simulated and trained quadruped locomotion policies using the Motion Imitation framework in PyBullet
- Validated a 1:16 multi-stage planetary gearbox producing torque outputs exceeding 30 Nm

Co-founder, Blufos

(Aug 2020 - Dec 2022)

- Led strategic B2B sales of 3D printers and rapid prototyping solutions
- Developed Python and LabVIEW tools for automated testing, sensor calibration, real-time telemetry
- Integrated OctoPrint API for remote 3D printer management and job scheduling, improving utilization

PROJECTS

MINIMAL ROBOT DOG - Graduate Thesis

- Researched the impact of **contact sensing and ground reaction force** emulation on linear control policy performance using Augmented Random Search in PyBullet
- Performed **sim-to-real transfer** using ROS and force-sensitive resistors for real-world validation

SCHVAAN ROBOT DOG

- Built a 12-DOF quadruped robot with custom **cycloidal actuators** and implemented a Cartesian-space locomotion controller
- Evaluated actuator performance for torque output, back-drivability, and low backlash

HUMAN ROBOT INTERACTION USING POINTING GESTURES

- Developed a real-time gesture-based system using MediaPipe (Pose Estimation), YOLOv4, and OpenCV
- Implemented geometric calculations to **estimate ground intersection points** based on body landmarks for enhanced spatial interaction

PUBLICATIONS

- Publication: (ICCAR 2025- Accepted): [Ground contact and reaction force sensing for linear policy control of quadruped robot](#)
- An article about the Schvaan robot was published in a leading National newspaper. - [Link](#)